Virtualization Technology

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Virtualization : An Introduction

- Virtualization is a large umbrella of technologies and concepts that are meant to provide an abstract environment-whether virtual hardware or an operating system.
- This technology allows to create multiple simulated environments or dedicated resources from a single, physical hardware system.
- Virtualization enables multiple operating systems to run on the same physical platform (running Windows OS on top of virtual machine, which itself is running on Linux OS).
- Virtualization provides a great opportunity to build elastically scalable systems, which are capable of provisioning additional capability with minimum costs.

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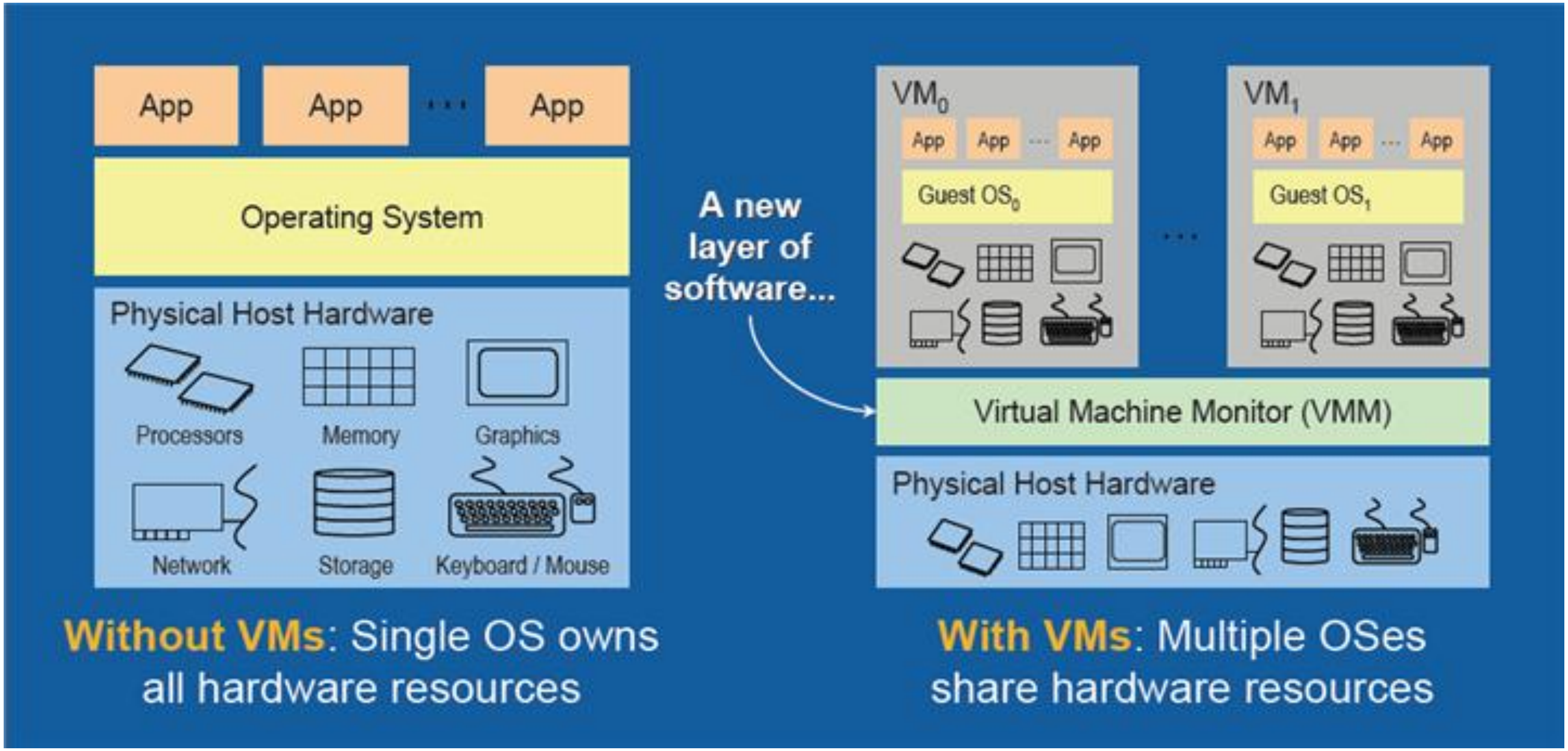
Virtualization : An Introduction



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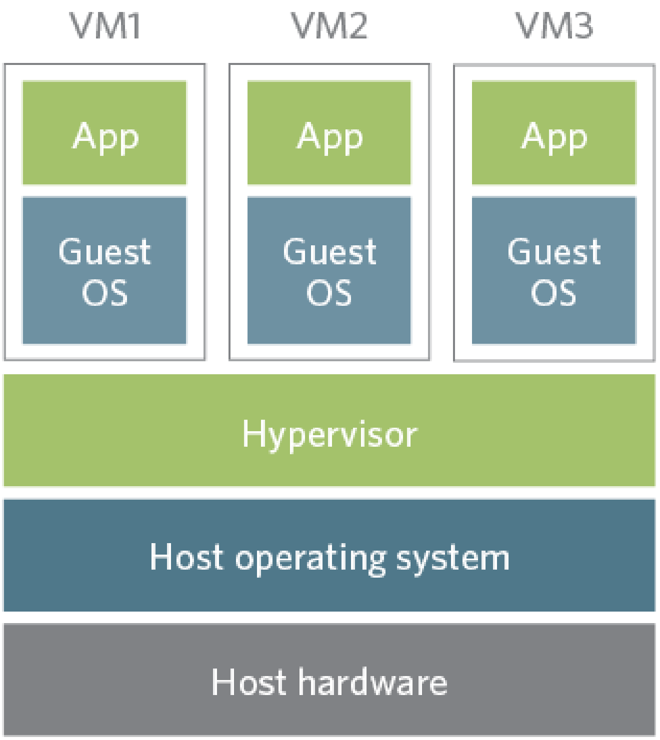
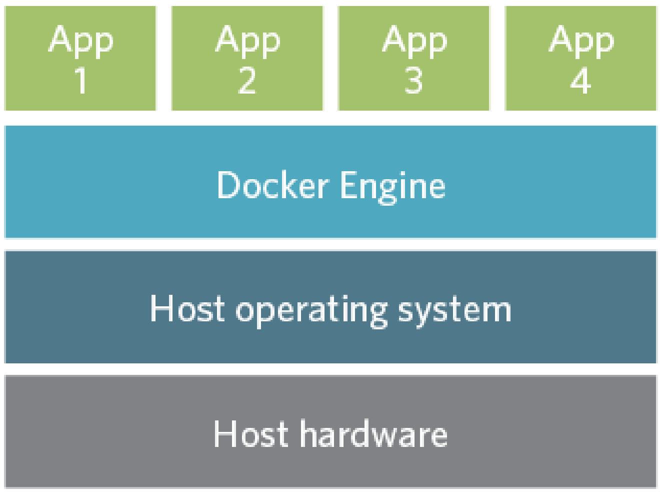
Virtualization : An Introduction

Virtualization Vs. Container



Virtualization



Container

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Virtualization : An Introduction

Virtualization Vs. Container

- The difference between a container versus a VM is primarily in the location of the virtualization layer and the way that operating system resources are used.
- VMs rely on a hypervisor which is normally installed over the actual physical system hardware.
- Once the hypervisor layer is installed, VM instances can be provisioned and each VM can then receive its own unique operating system and workload (application).
- VMs are fully isolated from one another -- no VM is aware of, or relies on, the presence of another VM on the same system.
- Issues i.e. malware attack, application crashes impact only the affected VM.

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Virtualization : An Introduction

Virtualization Vs. Container

- With containers, a host operating system is installed on the system first, and then a container layer.
- VMs rely on a hypervisor which is normally installed over the actual physical system hardware.
- Over the installed container layer, container instances can be provisioned and applications can be deployed within the containers.
- Every containerized application **shares the same underlying operating system** -- the single host OS.
- Containers are regarded as more resource-efficient than VMs as the additional resources needed for each OS is eliminated.
- The single OS presents a **single point of failure** for all containers.

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Significance for Adapting Virtualization

- Virtualization technologies have gained a renewed interest due to the confluence of different phenomena :--
    - Increased Performance and Computing Capacity
    - Underutilized Hardware and Software Resources
    - Lack of Space
    - Green Initiatives
    - Rise of Administrative Costs

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Characteristics of Virtualized Environments

- Virtualized environment consist of three major components: guest, host, and virtualization layer.
  - The **guest** represents the system component that interacts with the virtualization layer rather than the host.
  - The **host** represents the original environment where the guest is supposed to be managed.
  - **Virtualization layer** is responsible for recreating the same or a different environment where the guest will operate.

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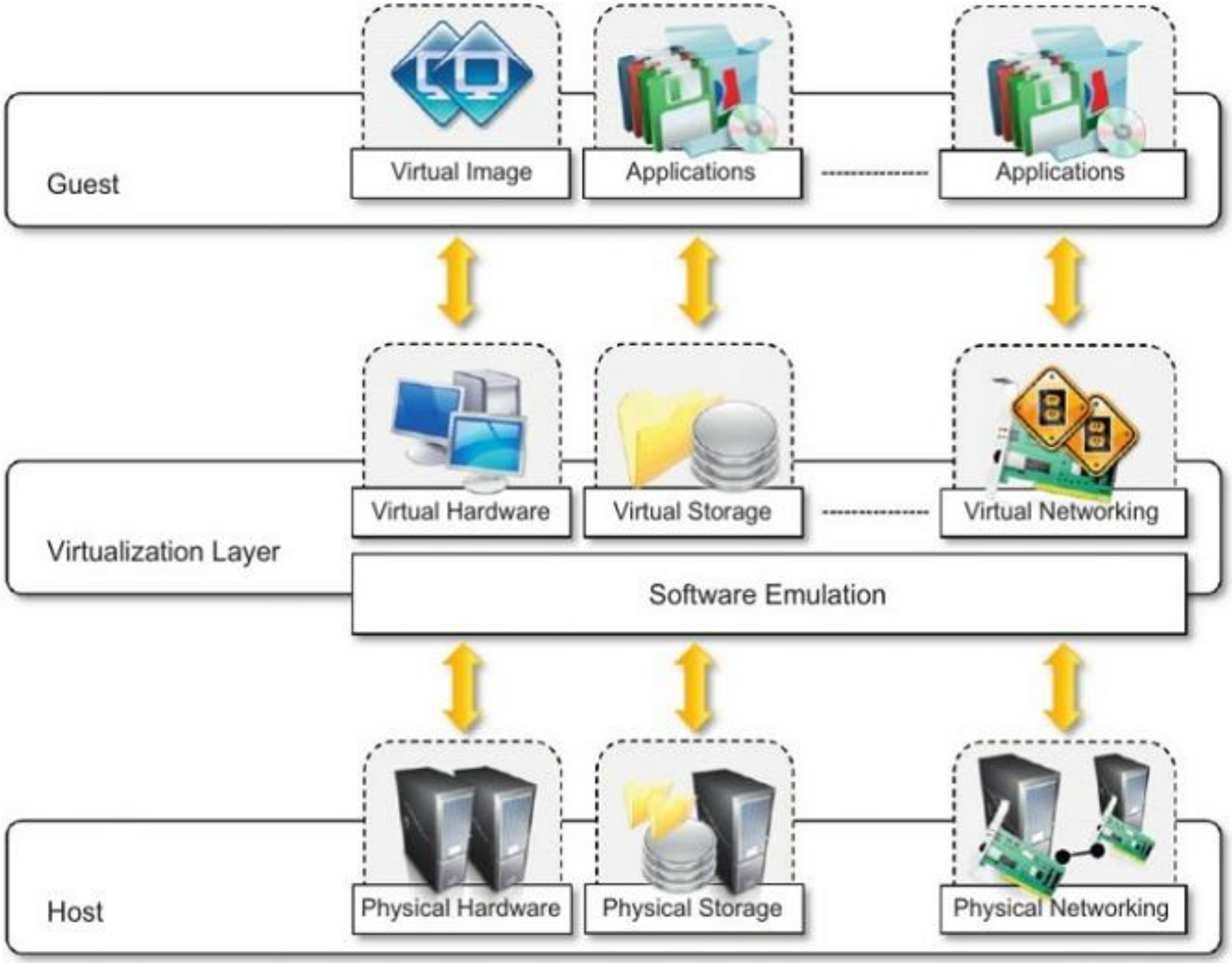
Virtualization Reference Model



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Advantages of Virtualization

- **Increased Security**
  - Ability to control the execution of the guest.
  - Guest is executed in an emulated environment.
  - VMM controls and filters the activity of the guest.
  - VM instances are isolated from each other.
- **Managed Execution**
  - **Sharing** -- Creating separate environment within the same host.
  - **Aggregation** -- A group of separate host can be tied together and represented as single virtual host.
  - **Emulation** -- Controlling and tuning the environment exposed to guest.
  - **Isolation** -- Complete separate environment for guests.

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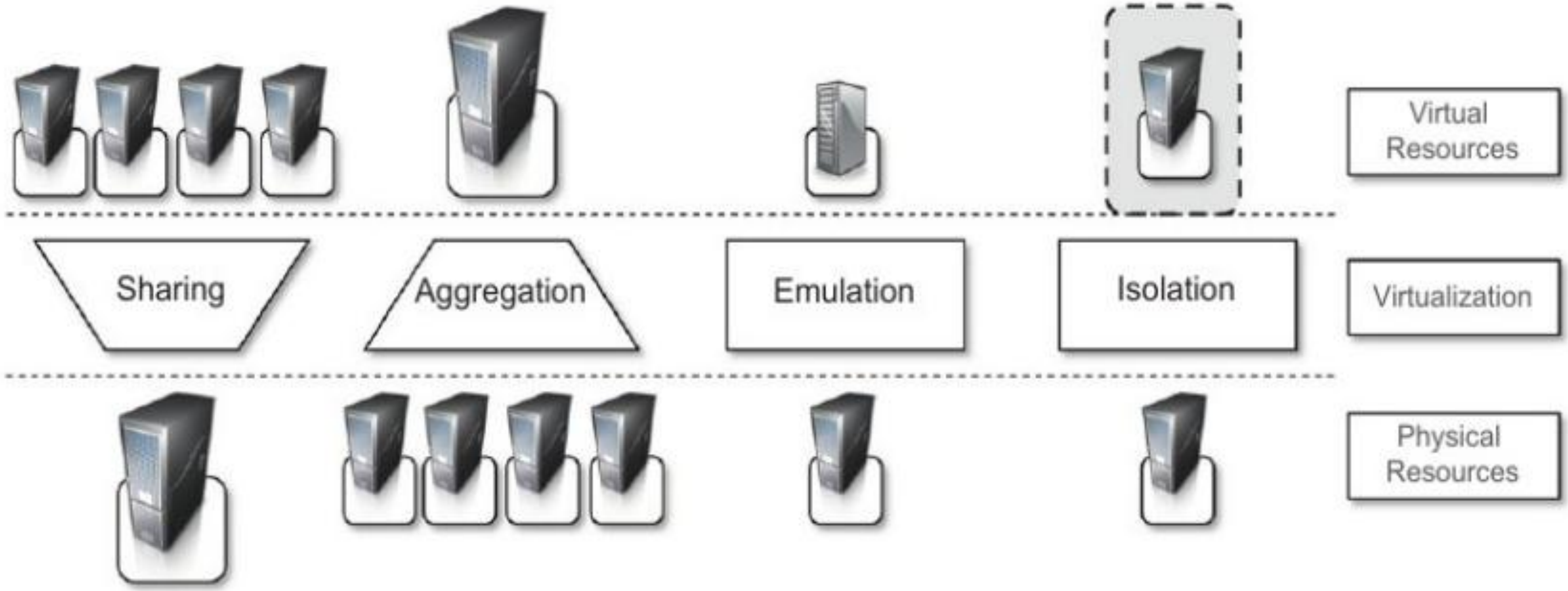
Managed Execution



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Advantages of Virtualization

- **Performance Tuning** -- Control the performance of guest.
- **Virtual Machine Migration** -- Move virtual image into another machine.
- **Portability** -- Virtual image can be safely moved and executed on top of different virtual machines.

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Taxonomy of Virtualization Techniques

- Virtualization is mainly used to emulate execution environment, storage and networks.
- Execution environment is classified into :-
  - **Process Level --** Implemented on top of an existing operating system.
  - **System level --** Implemented directly on hardware and do not or minimum requirement of existing operating system.

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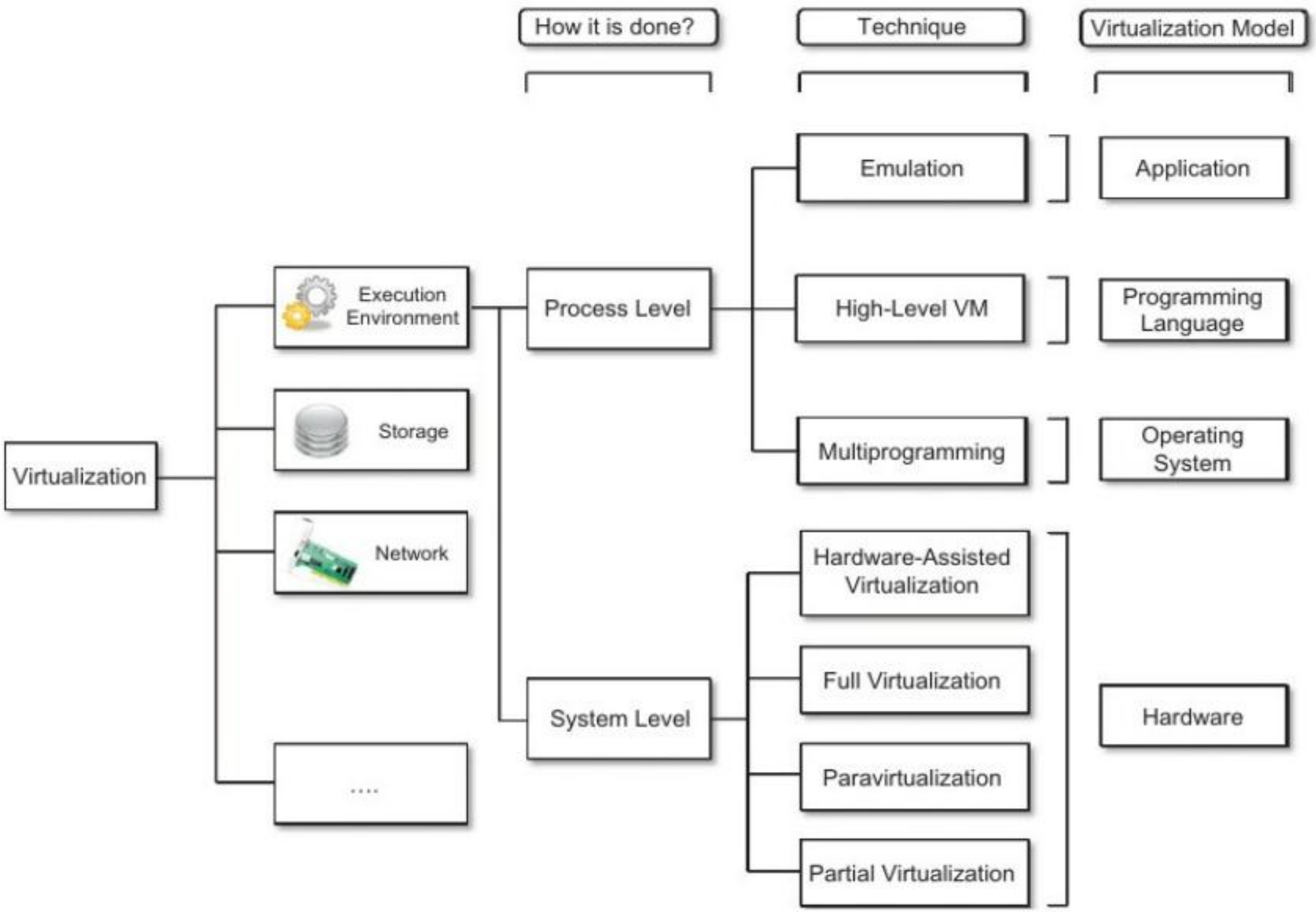
Taxonomy of Virtualization Techniques



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Execution Virtualization

- Execution virtualization includes all those techniques whose aim is to emulate an execution environment that is separate from the one hosting the virtualization layer.
- Execution virtualization can be implemented directly on top of the hardware, by the operating system, an application, or libraries dynamically or statically linked against an application image.
- Virtualizing an execution environment at different levels of the computing stack requires an reference model (**Machine Reference Model**) that defines the interfaces between the levels of abstractions, which hide implementation details.

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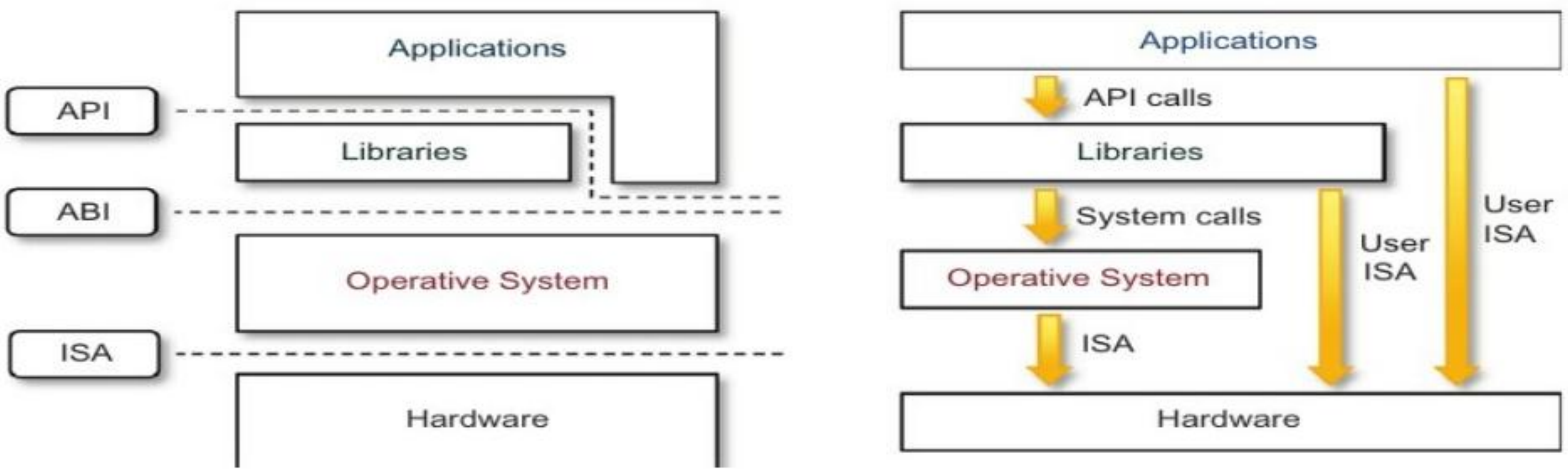
Machine Reference Model



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Hardware Level Virtualization

- It is a virtualization technique that provides an abstract execution environment in terms of computer hardware, on top of which a guest OS can be run.
- In this model, the guest is represented by OS, the host by physical computer hardware, the virtual machine by its emulation, and VMM by its hypervisor.
- The hypervisor is generally a program, or a combination of software and hardware, that allows the abstraction of the underlying physical hardware.
- There are two major types of hypervisor i.e. Type I and Type II

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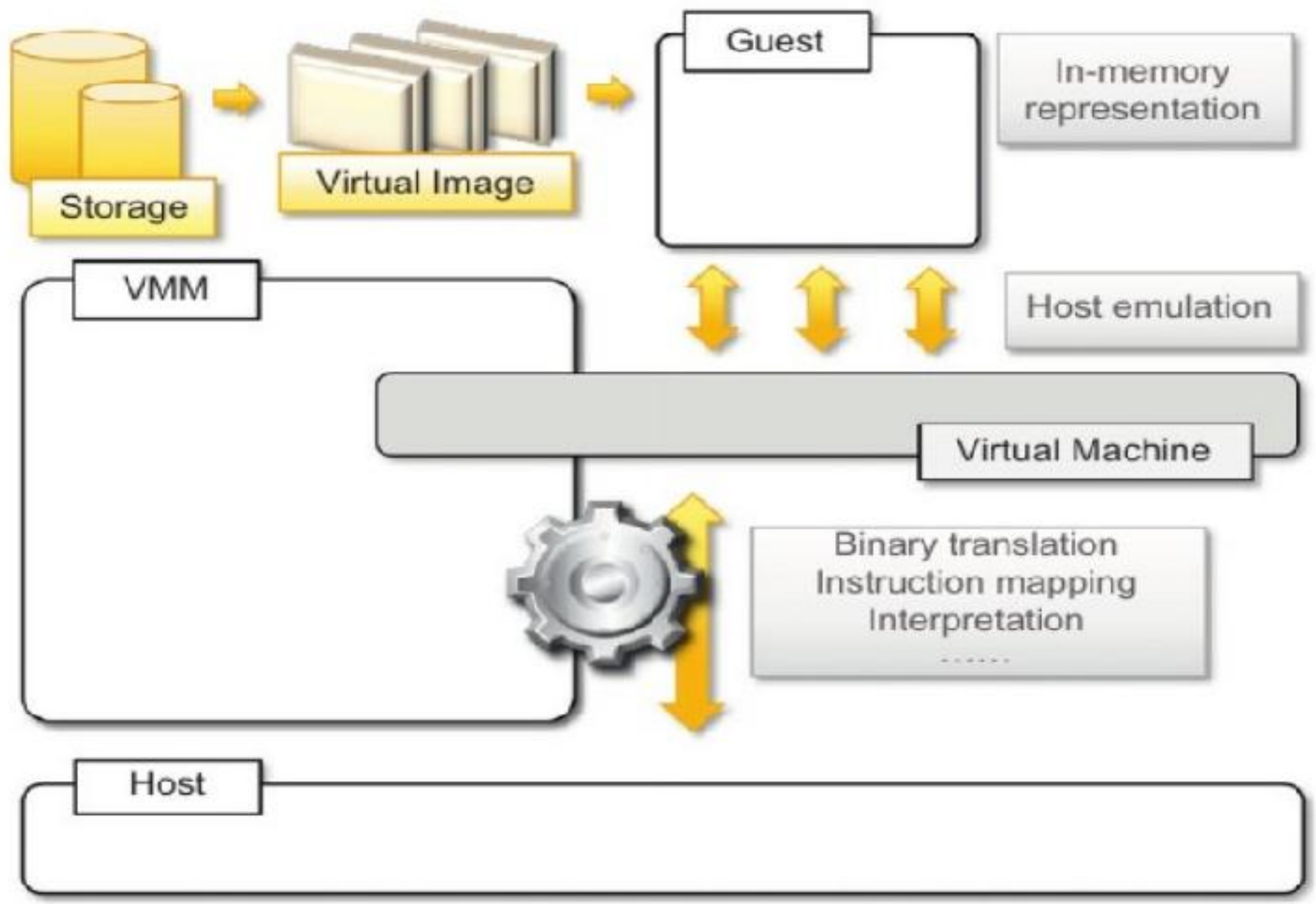
Hardware Level Virtualization



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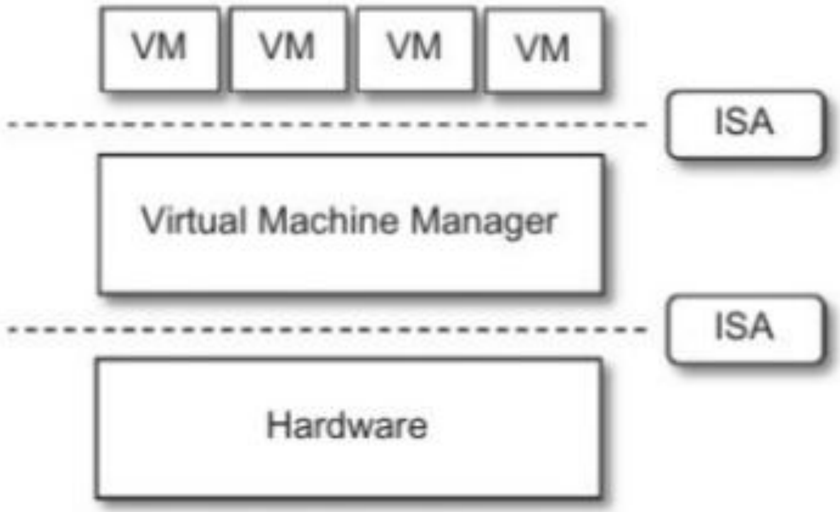
Type I Hypervisor

- It runs directly on top of the hardware and takes the place of OS.
- Directly interact with the ISA interface exposed by the underlying hardware.



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Type II Hypervisor

- This hypervisor requires the support of an OS to provide virtualization services.
- It consists of programs which are managed by the OS.
- The OS interact with the hypervisor through the ABI.
- The hypervisor emulate the ISA of virtual hardware for the guest OS.

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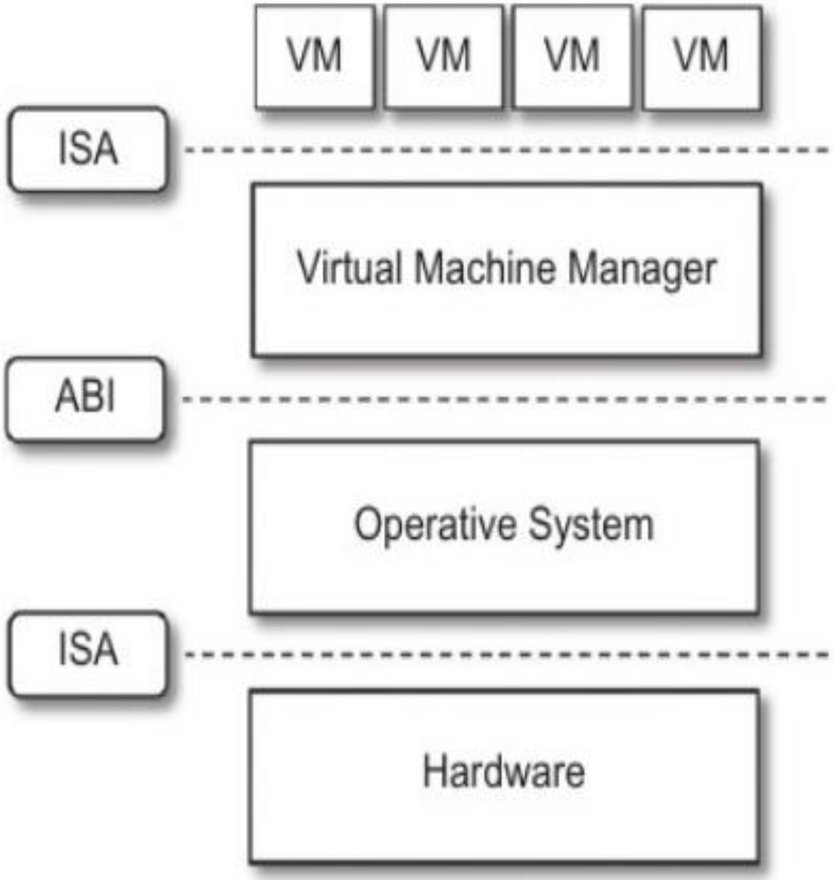
Type II Hypervisor



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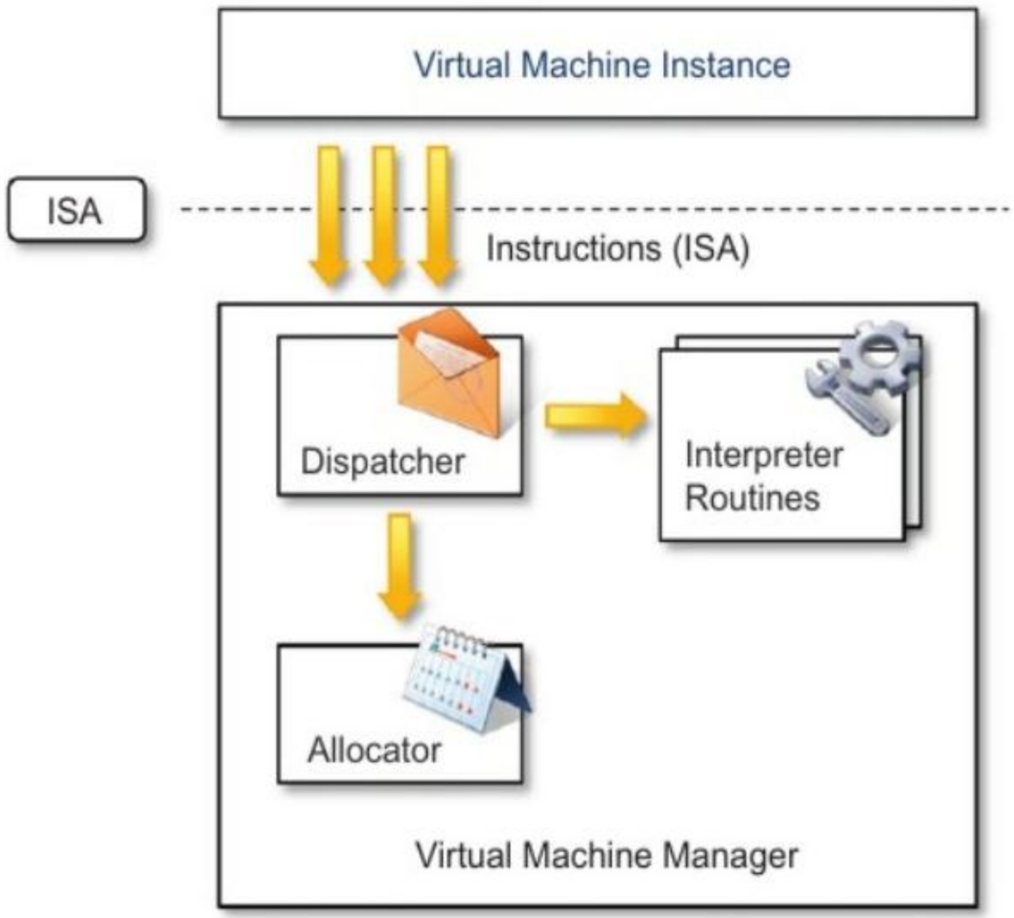
Hypervisor Reference Architecture



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Hardware Virtualization Techniques

- CPU installed on the host is only one set, but each VM that runs on the host requires there one CPU.
- It means CPU needs to be virtualized, and is performed by a hypervisor.

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Types of Hardware Level Virtualization

- **Hardware assisted virtualization:** In this, H/W provides architectural support for building a VMM able to run a guest OS in complete isolation.
- **Full virtualization :**
  - Ability to run program (OS) on top of a virtual machine and without any modification.
  - VMM requires complete emulation of the entire underneath hardware.
- **Para virtualization:**
  - This is not a transparent virtualization solution that allows implementing thin virtual machine manager.
  - Expose software interface to the virtual machine that is slightly modified by the host.
  - Guest OS need to be modified.
- **Partial virtualization :**
  - Partial emulation of the underlying hardware.
  - Not allow the complete execution of the guest OS in complete isolation.

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Operating System Level Virtualization

- It offers the opportunity to create different and separate execution environments for applications that are managed concurrently.
- No VMM or hypervisor is required.
- Virtualization is done within a single OS.
- OS kernel allows for multiple isolated user space instances .

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Programming Language Level Virtualization

- Is mostly used for achieving ease of deployment, managed execution, and portability across different platforms and OS.
- It consists of a virtual machine executing the byte code of a program, which is the result of the compilation process.
- Produce a binary format representing the machine code for an abstract architecture.
- Provide uniform execution environment across different platforms.

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Application Level Virtualization

- Its a technique allowing applications to run in runtime environments that do not natively support all the features required by such applications.
- Applications are not installed in the expected runtime environment, but run as if they are.

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Other Types of Virtualization

- Storage
- Network
- Desktop

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Disadvantages of Virtualization

- Performance degradation
- Inefficiency and degraded user experience
- Security holes and new threats
- Migration Issues

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Reference

1. T. Velte, A. Velte and R. Elsenpeter-Cloud Computing: A Practical Approach, Mc Graw Hill, India.
2. R. Buyya, C. Vecchiola and S. T. Selvi, Mastering Cloud Computing, Mc Graw Hill, India.

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QUESTIONS

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THANK YOU

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